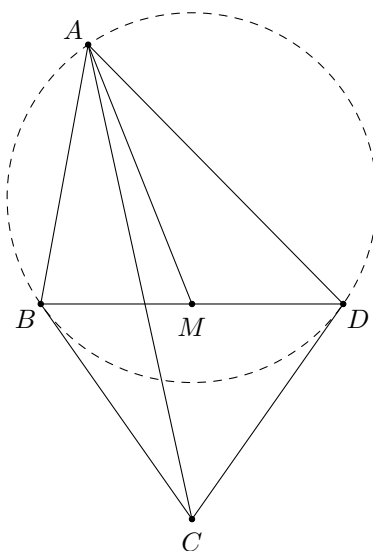


Problem 57 (Mathematical Reflections). Let $ABCD$ be a convex quadrilateral such that the sides BC and CD have equal length and $2\angle A + \angle C = 180^\circ$. Let M be the midpoint of BD . Prove that $\angle MAD = \angle BAC$.

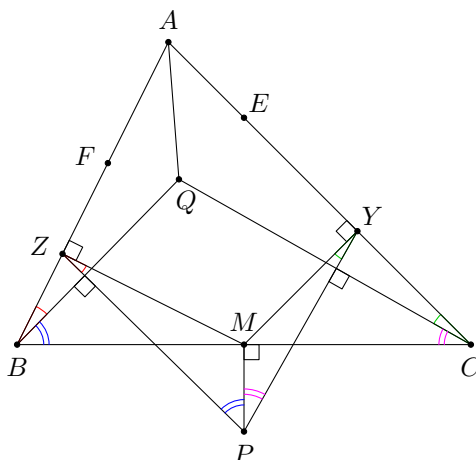


Solution. Note that

$$\angle CDB = \angle CBD = \frac{180^\circ - \angle BCD}{2} = \frac{2\angle BAC}{2} = \angle BAC.$$

So BC and CD are both tangent to the circumcircle of $\triangle ABD$. Hence, C is the A -symmedian point of $\triangle ABD$. This immediately implies $\angle CAB = \angle MAD$. $\square$

Problem 58. Let ABC be a triangle and BE, CF be the altitudes from B, C respectively. Let Y and Z be the midpoints of CE and BF respectively. The point P is an arbitrary point on the perpendicular bisector of BC . Let Q be the intersection of the line through B perpendicular to PZ and the line through C perpendicular to PY . Prove that AQ is the symmedian of $\triangle ABC$.



Solution. First of all, let M be the midpoint of BC . Note that $MZ \perp AB$, $MY \perp AC$ and $BC \perp MP$. Now the points E and F become useless. We only work on the case when P lies outside $\triangle ABC$, since the other case is similar.

By simple angle chasing (using the 5 right angles), we see that $\angle ABQ = \angle MZP$, $\angle MBQ = \angle ZPM$, $\angle ACQ = \angle MYP$ and $\angle QCM = \angle MPY$. Now, we use the angular form of Ceva's theorem and sine formula. We see that

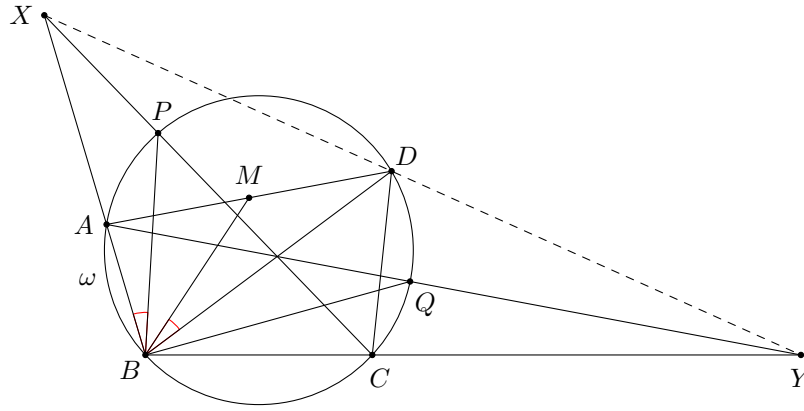
$$\begin{aligned} & \frac{\sin \angle BAQ}{\sin \angle CAQ} \cdot \frac{\sin \angle ACQ}{\sin \angle QCM} \cdot \frac{\sin \angle MBQ}{\sin \angle QBA} = 1 \\ \Rightarrow & \frac{\sin \angle BAQ}{\sin \angle CAQ} \cdot \frac{\sin \angle MYP}{\sin \angle MPY} \cdot \frac{\sin \angle ZPM}{\sin \angle MZP} = 1 \\ \Rightarrow & \frac{\sin \angle BAQ}{\sin \angle CAQ} \cdot \frac{MP}{MY} \cdot \frac{ZM}{MP} = 1. \end{aligned}$$

Therefore,

$$\frac{\sin \angle BAQ}{\sin \angle CAQ} = \frac{MY}{MZ} = \frac{MC \cdot \sin \angle ACB}{MB \cdot \sin \angle ABC} = \frac{\sin \angle ACB}{\sin \angle ABC} = \frac{AB}{AC},$$

which is sufficient to show that AQ is the A -symmedian of $\triangle ABC$. $\square$

Problem 59 (Thailand TST 2015). Let $ABCD$ be a quadrilateral inscribed in a circle ω . The symmedians through B of $\triangle ABD$ and $\triangle CBD$ intersect ω again at points P and Q , respectively. Lines CP and AB intersect at X , and lines AQ and BC intersect at Y . Prove that the points X, D, Y are collinear.

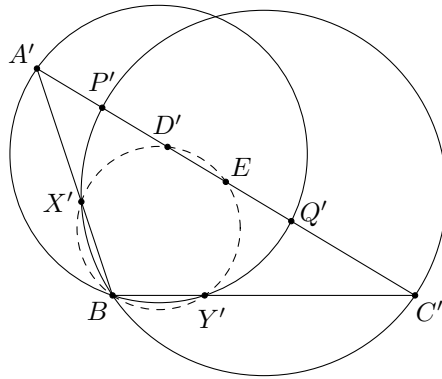


Solution. Invert the figure about a circle centred at B . For each point Z , let Z' be the image of Z under the inversion. Note that $(ABCD)$ is mapped to a line $A'C'$. Also, X' is the intersection of line BA' and $(C'P'B)$, while Y' is the intersection of line BC' and $(A'Q'B)$.

Let M be the midpoint of AD . Due to the inversion, $\triangle BDA \sim \triangle BA'D'$, so we have $\triangle BDM \sim \triangle BA'M'$. Note that under inversion, $\angle ABP = \angle A'BP'$. Hence,

$$\angle A'BP' = \angle ABP = \angle DBM = \angle A'BM'.$$

Since P' and M' are both points on line $A'C'$, we see that $P' = M'$, i.e. P' is the midpoint of $A'D'$. Similarly, Q' is the midpoint of $C'D'$.



Finally, the conclusion is that B, X', D', Y' are concyclic.

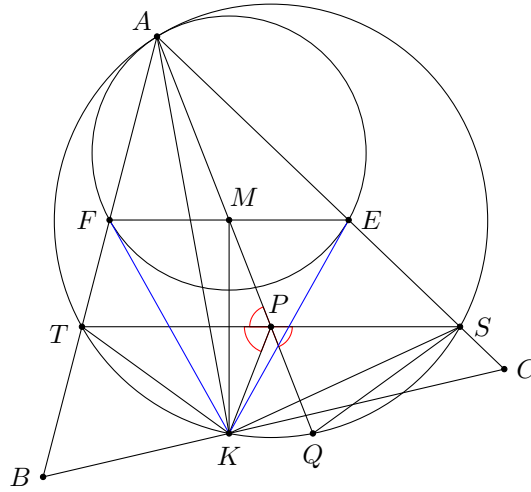
To prove this, let $AP' = P'D' = x$ and $C'Q' = Q'D' = y$. Suppose $(BX'D')$ intersects $A'C'$ again at E . Then

$$A'E = \frac{A'E \cdot A'D'}{2x} = \frac{A'X' \cdot A'B}{2x} = \frac{A'P' \cdot A'C'}{2x} = \frac{(x)(2x + 2y)}{2x} = x + y.$$

Hence, E is the midpoint of $A'C'$. Similarly, D', E, Y', B are concyclic. It follows that B, X', D', E, Y' are concyclic. $\square$

Problem 60 (IMO shortlist 2014 G6). Let ABC be a fixed acute-angled triangle. Consider some points E and F lying on the sides AC and AB , respectively, and let M be the midpoint of EF . Let the perpendicular bisector of EF intersect the line BC at K , and let the perpendicular bisector of MK intersect the lines AC and AB at S and T , respectively. We call the pair (E, F) *interesting*, if the quadrilateral $KSAT$ is cyclic.

Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. Prove that $\frac{E_1E_2}{AB} = \frac{F_1F_2}{AC}$.



Solution. The solution consists of two steps.

Step 1: Symmedian

Extend AM to meet ST and $(ATKS)$ at P and Q respectively. In this step, we can ignore B, C .

Note that PT is the perpendicular bisector of MK , so $MP = PK$ and PT is the angle bisector of $\angle MPK$. Thus

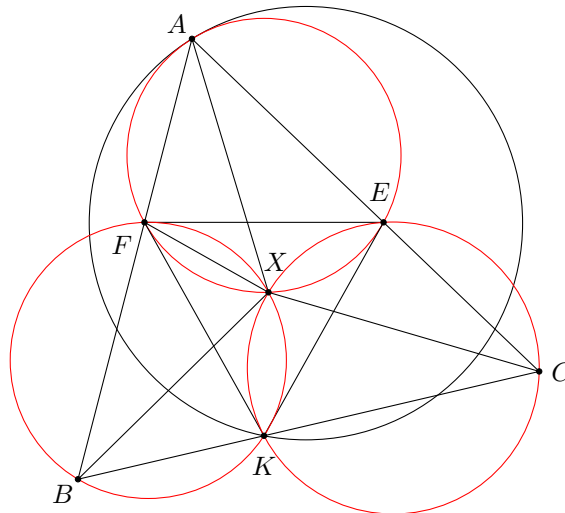
$$\angle KPT = \angle MPT = \angle QPS.$$

Also, since EF is parallel to ST and $MF = ME$, we have $PT = PS$. Hence, by symmetry (reflectional symmetry along perpendicular bisector of ST), we see that $TK = SQ$. Hence, $\angle EAM = \angle FAK$, meaning that AK is the A -symmedian of $\triangle AEF$.

If $AE \neq AF$, the A -symmedian of $\triangle AEF$ and the perpendicular bisector of EF has a unique intersection point. We know this point is K . Therefore, the two tangents of (AEF) at E and F intersect at K .

The same result holds if $AE = AF$. Indeed, in that case AK is a diameter of $(ATKS)$. This implies F, T, K, M are concyclic. Then the result follows from

$$\angle TFK = \angle TMK = \angle MKT = \angle AKT = \angle AST = \angle AEF.$$



Step 2: Spiral similarity

Let X be the Miquel point of $\triangle EFK$ with respect to $\triangle ABC$. Then

$$\angle CBX = \angle KBX = \angle KFX = \angle FAX = \angle BAX.$$

Similarly, $\angle BCX = \angle CAX$. This shows X is the A -Humpty point of $\triangle ABC$. In particular, it is independent from E and F .

Suppose that the pairs (E_1, F_1) and (E_2, F_2) are interesting. From above, (AE_1F_1) and (AE_2F_2) meet at A and X . Therefore, X is the centre of spiral similarity sending E_1E_2 to F_1F_2 . It follows that

$$\frac{E_1E_2}{F_1F_2} = \frac{XE_1}{XF_1} = \frac{d(X, AC)}{d(X, AB)} = \frac{AB}{AC}.$$

Here, $d(Z, \ell)$ means the distance from Z to ℓ . The last equality follows from the fact that X lies on the A -median of $\triangle ABC$. $\square$